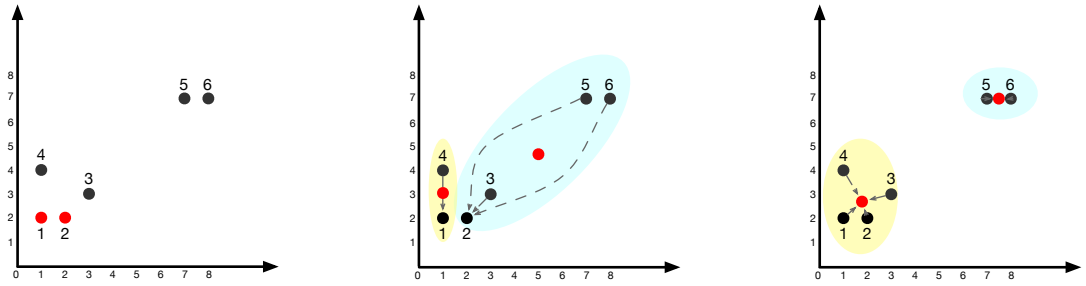
XIAMEN UNIVERSITY MALAYSIA

Marking Scheme

1. The K-means method will produce the following clusters after two iterations:



Initial status First iteration Second iteration

(a) After the first iteration:

Cluster 1: {p1, p4}

Cluster 2: {p2, p3, p5, p6}

(b) After the first iteration:

Centroid of cluster 1: $(\frac{1+1}{2}, \frac{2+4}{2}) = (1, 3)$

Centroid of cluster 2: $(\frac{2+3+7+8}{4}, \frac{2+3+7+7}{4}) = (5, \frac{19}{4})$

(c) After the second iteration:

Cluster 1: {p1, p2, p3, p4}

Cluster 2: {p5, p6}

(d) Since the centroids are fixed after the second iteration, we obtain the final centroids:

Centroid of cluster 1: $(\frac{1+2+3+1}{4}, \frac{2+2+3+4}{4}) = (\frac{7}{4}, \frac{11}{4})$

Centroid of cluster 2: $(\frac{7+8}{2}, \frac{7+7}{2}) = (\frac{15}{2}, 7)$